

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

ECCC3073/UCCC3073 DATA SCIENCE

THURSDAY, 26 SEPTEMBER 2024

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the **FIRST (1)** question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

ECCC3073/UCCC3073 DATA SCIENCE**SECTION A (Answer ALL questions)**

- Q1. (a) Complete Pet Care Ltd. provides private healthcare for domestic pets throughout Malaysia. The company has more than 100 surgeries and decided to distribute its operations to 12 different states in the country. The following are the tables stored in the company's database.

Owner	(ownerNo, oName, oAddress, oTelNo)
Pet	(petNo, petName, petDesc, petSex, petDOB, dateRegistered, ownerNo, surgeryNo)
Staff	(staffNo, sName, sAddress, sTelNo, sex, DOB, position, salary, surgeryNo)
Surgery	(surgeryNo, surgeryAddress, surgeryTelNo, mgrStaffNo, areaNo)
Prescription	(prescDateTime, petNo, medNo, prescStaffNo, adminMethod, unitsPerDay, startDate, finishDate)
Medication	(medNo, medName, description, dosage, cost)
Area	(areaNo, areaName)

- (i) Plot a conceptual model to illustrate relationships between entities. (7 marks)
- (ii) Outline primary key(s) and foreign key(s) for each entity. (8 marks)
- (b) Table 1.1 shows the records of 4 students with their CGPA, year of study, and age, respectively.

Table 1.1

Student name	CGPA	Age	Year of study
Mohammad	3.2	19	2
Krishnan Muthu	3.0	20	?
Jennifer Lau	2.7	18	1
Lim Xi Feng	3.5	20	3
Aisha	2.9	20	1

- (i) Calculate Euclidean distance between Krishnan Muthu and the other 3 students based on CGPA and Age, and recommend **TWO (2)** nearest neighbors of Krishnan Muthu. Round your answer to the nearest 2 decimal points. (8 marks)
- (ii) Using the answer of Q1(b)(i), estimate Krishnan Muthu's years of study by using the average of the year of study of 2 nearest neighbors. (2 marks)
- [Total : 25 marks]

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- Q2. A pharmaceutical company claims that their new medication reduces the average recovery time for patients with a specific condition from the current average of 10 days. To test this claim, clinical trial based on a random sample of 5 patients are conducted, and the average recovery time for patients with the new medication is 8 days. Table 2.1 shows the recovery time for 5 patients.

Table 2.1

Patient	Recovery time in Days
1	5
2	10
3	8
4	9
5	10

Conduct a hypothesis test to determine if there is sufficient evidence to support the company's claim at a 5% significance level.

- (a) Determine null and alternative hypotheses. (2 marks)
- (b) Determine the following.
- (i) Test result. Round your answer to the nearest 3 decimal points. (6 marks)
- (ii) p-value. Round your answer to the nearest 3 decimal points. (2 marks)
- (c) Using the answers of Q2(b), justify whether the null hypothesis of Q2(a) can be rejected. (2 marks)
- (d) Assume that the random sample of Q2 is increased to 200 patients and the population standard deviation is 3.5 days. Calculate the following:
- (i) Test result. Round your answer to the nearest 3 decimal points. (4 marks)
- (ii) p-value. Round your answer to the nearest 3 decimal points. (2 marks)
- (iii) Upper- and lower bounds. Round your answer to the nearest 3 decimal points. (4 marks)
- (e) Using the computed results of Q2(d), justify whether the null hypothesis of Q2(a) can be rejected. (3 marks)
- [Total : 25 marks]

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- Q3. Given the following dataset of total number of sunshine hours and amount of rainfall (mm) in Kampar for the second half of year 2023.

Table Q3.1

Month	Sunshine (hours)	Rainfall (mm)
July	294	128
August	293	152
September	264	275
October	262	378
November	227	433
December	270	311

Assume a linear model is used to model the problem.

- (a) Determine the best-fit line for the data. Round your answer to the nearest 2 decimal points. (10 marks)
- (b) Comment the calculated regression equation in Q3(a). (3 marks)
- (c) Determine the Pearson's correlation coefficient, r , between the total number of sunshine hours and amount of rainfall (mm) and use it to comment on the relationship between sunshine and rainfall. Round your answer to the nearest 2 decimal points. (4 marks)
- (d) Apply the regression line of Q3(a) to determine the amount of sunshine (in hours) when Kampar experiencing a heavy rain pour of 200 mm in a month. Comment on the reliability of your prediction. (4 marks)
- (e) Calculate the coefficient of determination, R^2 , for the predicted regression line of Q3(a) and use it to comment on the reliability of your prediction in Q3(d). Round your answer to the nearest 2 decimal points. (4 marks)

[Total : 25 marks]

ECCCC3073/UCCCC3073 DATA SCIENCE**SECTION B (Choose only ONE question)**

- Q4. You have developed a binary classification model to predict the likelihood of a patient having a certain medical condition. The model was tested on a dataset of 200 patients, producing the following confusion matrix:

Table 4.1

	Predicted healthy	Predicted disease
Actual healthy	140	10
Actual disease	30	20

- (a) Calculate the baseline performance. (2 marks)
- (b) Define true positive (TP), true negative (TN), false positive (FP) and false negative (FN) in the context of medical diagnosis. (4 marks)
- (c) Using the definition of Q4(b), calculate the following evaluation measures. Round your answer to the nearest 2 decimal points.
- (i) Classification accuracy rate. (1 mark)
 - (ii) Precision. (1 mark)
 - (iii) Recall. (1 mark)
 - (iv) Specificity. (1 mark)
 - (v) F_1 score. (2 marks)
- (d) Using the computed results of Q4(a) and Q4(c), justify whether you would propose the model to the hospital. (7 marks)
- (e) Suggest **TWO (2)** ways to overcome the issues you have observed in Q4(d). (6 marks)
- [Total : 25 marks]

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- Q5. You have developed two classification models, Model A and Model B, for medical diagnostic. Table 5.1 shows the prediction results of the models on the test data.

Table 5.1

N = 1050		Predicted values			
		Model A		Model B	
		Healthy	Cancer	Healthy	Cancer
Target/ Actual	Healthy	850	50	860	40
	Cancer	30	120	40	110

- (a) Calculate the baseline performance. (2 marks)
- (b) Calculate the following evaluation measures for both models. For each measure, identify the best model. Round your answer to the nearest 2 decimal points.
- (i) Misclassification rate. (3 marks)
 - (ii) Recall. (3 marks)
 - (iii) Specificity. (3 marks)
 - (iv) Precision. (3 marks)
- (c) Using the computed results of Q5(a) and Q5(b), justify whether you would propose Model A to the hospital. (3 marks)
- (d) Assume that Model A uses equal cost and Model B uses unequal cost where the cost for false positive ($Cost_{FP}$) is 5 and the cost for true positive ($Cost_{TP}$) is -15.
- (i) Determine the model profit per record for both models. (3 marks)
 - (ii) Justify whether the unequal cost model is better than the baseline performance and the equal cost model. (5 marks)
- [Total : 25 marks]

APPENDIX

Proximity measures	
Euclidean distance $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - x)^2 + (y_i - y)^2}$ Manhattan distance $d(x, y) = \sum_{i=1}^n x_i - y_i $ $d(x, y) = \sum_{i=1}^n x_i - x + y_i - y $ Supremum distance $d(x, y) = \max(x_i - y_i)$ $d(x, y) = \max(x_i - x , y_i - y)$	Hamming distance $d(a, b) = (a \oplus b)$ Simple matching coefficient $SMC = \frac{\text{num of matching attriute values}}{\text{num of attributes}}$ Jaccard coefficient $J = \frac{\text{num of matching presences}}{\text{num of attributes not involved in 00 matches}}$ Cosine similarity $\cos(x, y) = \frac{x \cdot y}{ x y }$
Regression	
Straight line $\hat{y} = \beta_0 + \beta_1 x$ Coefficients $\hat{\beta}_1 = \frac{p \sum xy - \sum x \sum y}{p \sum x^2 - (\sum x)^2}$ $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ Residuals $\hat{\varepsilon} = y - \hat{y}$ $s^2 = \frac{\sum \hat{\varepsilon}^2}{p - 2}$ $s = \sqrt{s^2}$ t value $t = \frac{\hat{\beta}}{SE}$	Coefficient of Determination $R^2 = \left[\frac{p \sum xy - \sum x \sum y}{\sqrt{(p \sum x^2 - (\sum x)^2) * (p \sum y^2 - (\sum y)^2)}} \right]^2$ Pearson correlation $R = \sqrt{R^2}$ Standard errors of coefficients $SE(\hat{\beta}_1) = \frac{s}{\sqrt{\sum (x - \bar{x})^2}}$ $SE(\hat{\beta}_0) = s \sqrt{\frac{1}{p} + \frac{\bar{x}^2}{\sum (x - \bar{x})^2}}$
Naïve Bayes	
Bayes' rule $P(H E) = \frac{P(E H)P(H)}{P(E)}$ Probability density function $pdf(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\bar{x})^2}{2\sigma^2}}$	Laplace correction $\frac{x_i + 1}{c + \alpha}$

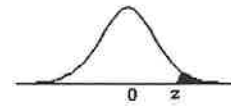
Association Rules	
Association rule $X \rightarrow Y$ Support $s = \frac{freq(X,Y)}{N}$	Confidence $c = \frac{freq(X,Y)}{freq(X)}$ Lift $l = \frac{Confidence(X,Y)}{Support(X)}$

Evaluation metrics	
Classification accuracy $acc = \frac{TN + TP}{TN + TP + FN + FP}$ Misclassification rate $1 - acc$ Sensitivity / Recall $recall = \frac{TP}{TP + FN}$ Specificity $spec = \frac{TN}{TN + FP}$ False discovery rate $FDR = \frac{FP}{FP + TP}$	Precision / Positive Predictive Value $PPV = \frac{TP}{TP + FP} = 1 - FDR$ F1 score $F_1 = \frac{2TP}{2TP + FP + FN}$ Overall model cost $FP.Cost_{FP} + TP.Cost_{TP}$ Model cost per case $\frac{Overall\ model\ cost}{N}$ Model profit per case $-model\ cost\ per\ case$

Miscellaneous	
Linear scaling for [a, b] $x' = (b - a) \frac{x - \min(x)}{\max(x) - \min(x)} + a$ z-score normalization $x' = \frac{x - \mu}{\sigma}$ Mean, $\bar{x}$ $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ Z-test $Z = \frac{\bar{x} - \mu}{SE_{\bar{x}}}$ t-test $t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$	Inverse linear scaling for [a, b] $x = (x' - a) * \frac{\max(x) - \min(x)}{b - a} + \min(x)$ Population standard deviation $\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$ Sample standard deviation $s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$ Standard error $SE = \frac{\sigma}{\sqrt{n}}$ Confidence intervals $CI = \bar{x} \pm z * SE$

ECCC3073/UCCC3073 DATA SCIENCE**Normal Curve Areas (Standard normal probability in right-hand tail)***for negative values of z, areas are found by symmetry* confidence level to Z-score

90%	1.645
95%	1.96
98%	2.33
99%	2.575



z	Second decimal place of z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

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t Distribution: Critical Values of t										
Degree of Freedom	1-tailed α	0.25	0.2	Significant level					0.01	0.005
				0.15	0.1	0.05	0.025	0.01		
2-tailed α	0.5	0.4	0.3	0.2	0.1	0.05	0.025	0.01	0.005	
1	1.000	1.376	1.963	3.078	6.314	12.706	31.821	63.657		
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925		
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841		
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604		
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032		
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707		
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499		
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355		
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250		
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169		
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106		
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055		
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012		
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977		
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947		
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921		
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898		
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878		
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861		
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845		
21	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831		
22	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819		
23	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807		
24	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797		
25	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787		
26	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779		
27	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771		
28	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763		
29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756		
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750		
31	0.682	0.853	1.054	1.309	1.696	2.040	2.453	2.744		
32	0.682	0.853	1.054	1.309	1.694	2.037	2.449	2.738		
33	0.682	0.853	1.053	1.308	1.692	2.035	2.445	2.733		
34	0.682	0.852	1.052	1.307	1.691	2.032	2.441	2.728		
35	0.682	0.852	1.052	1.306	1.690	2.030	2.438	2.724		
36	0.681	0.852	1.052	1.306	1.688	2.028	2.434	2.719		
37	0.681	0.851	1.051	1.305	1.687	2.026	2.431	2.715		
38	0.681	0.851	1.051	1.304	1.686	2.024	2.429	2.712		
39	0.681	0.851	1.050	1.304	1.685	2.023	2.426	2.708		
40	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704		
41	0.681	0.850	1.050	1.303	1.683	2.020	2.421	2.701		
42	0.680	0.850	1.049	1.302	1.682	2.018	2.418	2.698		
43	0.680	0.850	1.049	1.302	1.681	2.017	2.416	2.695		
44	0.680	0.850	1.049	1.301	1.680	2.015	2.414	2.692		
45	0.680	0.850	1.049	1.301	1.679	2.014	2.412	2.690		
46	0.680	0.850	1.048	1.300	1.679	2.013	2.410	2.687		
47	0.680	0.849	1.048	1.300	1.678	2.012	2.408	2.685		
48	0.680	0.849	1.048	1.299	1.677	2.011	2.407	2.682		
49	0.680	0.849	1.048	1.299	1.677	2.010	2.405	2.680		
50	0.679	0.849	1.047	1.299	1.676	2.009	2.403	2.678		
60	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660		
70	0.678	0.847	1.044	1.294	1.667	1.994	2.381	2.648		
80	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639		
90	0.677	0.846	1.042	1.291	1.662	1.987	2.368	2.632		
100	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626		
150	0.676	0.844	1.040	1.287	1.655	1.976	2.351	2.609		
200	0.676	0.843	1.039	1.286	1.653	1.972	2.345	2.601		
∞	0.675	0.842	1.036	1.282	1.645	1.960	2.327	2.576		